

ETHAN SUN

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Education

University of Waterloo – Waterloo, Ontario

Aug 2028 (Expected)

BCS, Honours Computer Science with Co-op

GPA: 3.8/4.0

President's Scholarship of Distinction

Relevant Coursework: Data Structures and Algorithms, Automated Testing, Build Automation

Skills

Languages: Python, TypeScript, JavaScript, C++, SQL, C, Java, HTML/CSS

AI / ML: scikit-learn, PyTorch, Amazon Bedrock, OpenAI API, Claude API, pandas, NumPy, TensorFlow

Technologies / Frameworks: Next.js, React, Node.js, Express.js, FastAPI, Supabase, PostgreSQL

Tools: Git, GitHub, Docker, AWS Lambda, Linux, Bash, CI/CD, Vercel

Experience

Skrypt Health

Toronto, Ontario

AI Engineer Intern (Part-Time)

Sep 2026 – Present

- Developing an ML-based call evaluation system using historical transcripts, call metadata, and outcomes to identify patterns in AI receptionist performance, predict unsuccessful calls, and classify common failure modes.
- Automated AI voice agent creation with the Retell and Claude APIs, replacing a manual ops workflow and cutting per-agent setup time by 65%+ (6 min → under 2 min).

Software Engineer Intern

May 2026 – Aug 2026

- Built a Python outbound audit system that scheduled **9600+** calls to **100+** clinics within configured time windows and categorized responses via LLM classification, tokenizing patient PII before model calls.
- Built TypeScript/**AWS Lambda** webhook pipelines processing **18K+** Retell call events (**200/day**), extracting transcripts and summaries and syncing them into internal clinic systems within **15** seconds of call end.
- Migrated primary storage from Airtable to Supabase (PostgreSQL) to support growing call volume, designing the schema and webhook ingestion for call, SMS, and agent records (**400** events/day)

Electrium Mobility

Sep 2025 – Dec 2025

Firmware Developer

Waterloo, Ontario

- Developed a C++ UART/BMS API for an ESP32-based embedded system, enabling reliable low-latency telemetry streaming of real-time pack voltage, current, temperature, and cell data at **50Hz**.
- Built a FreeRTOS-based **data pipeline** for BMS and VESC sensor ingestion, validation, and output, eliminating data loss during processing and improving system reliability under load.
- Implemented signal-processing logic to convert raw VESC motor RPM into real-time speed and distance metrics, delivering **±2%** accuracy in user-facing performance measurements.

Projects

Application Tracker [GitHub] [Link] | *Next.js, TypeScript, PostgreSQL, OpenAI, Google OAuth, Gmail API, Vercel*

- Built and deployed a multi-user application tracking platform on Vercel with application management, filtering, and Gmail-based status detection, achieving **96%** accuracy in identifying relevant emails.
- Implemented Google OAuth and Gmail API integration to securely read emails and classify messages with GPT-4o-mini as new applications or status updates, surfacing confidence scores and reasoning for user review.
- Built a server-side data layer with per-user row scoping, two-layer Zod validation, and audit logging for every change.

F1 Telemetry Driver Performance Analysis [GitHub] | *Python, pandas, FastF1, scikit-learn, NumPy, Matplotlib*

- Built an end-to-end Python data pipeline to ingest, clean, and preprocess F1 qualifying telemetry from the FastF1 API, engineering speed, throttle, and braking features across **4,446** driver comparisons.
- Trained a **Random Forest** regression model to predict lap-time deltas between driver pairs, achieving a MAE of **0.223s** and R^2 of **0.72** while outperforming a hand-tuned weighted baseline model.
- Designed the evaluation approach using **GroupShuffleSplit** to hold out **20%** of race weekends as the test set, preventing data leakage between sessions from the same event and providing a more reliable estimate of out-of-event generalization.